

Лабораторная работа №6

Управление процессами

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Цель работы

Получить навыки управления процессами операционной системы Linux.

Ход выполнения работы

```
rkogengar@rishardkogengar:~$ su
Password:
root@rishardkogengar:/home/rkogengar# sleep 3600 &
[1] 3447
root@rishardkogengar:/home/rkogengar# dd if=/dev/zerp of=/dev/null &
[2] 3500
dd: failed to open '/dev/zerp': No such file or directory
root@rishardkogengar:/home/rkogengar# dd if=/dev/zero of=/dev/null &
[3] 3514
[2] Exit 1 dd if=/dev/zerp of=/dev/null
root@rishardkogengar:/home/rkogengar# sleep 7200
^Z
[4]+ Stopped sleep 7200
root@rishardkogengar:/home/rkogengar# jobs
[1] Running sleep 3600 &
[3]- Running dd if=/dev/zero of=/dev/null &
[4]+ Stopped sleep 7200
root@rishardkogengar:/home/rkogengar# bg 4
[4]+ sleep 7200 &
root@rishardkogengar:/home/rkogengar# jobs
[1] Running sleep 3600 &
[3]- Running dd if=/dev/zero of=/dev/null &
[4]+ Running sleep 7200 &
root@rishardkogengar:/home/rkogengar# fg 1
sleep 3600
^C
root@rishardkogengar:/home/rkogengar# fg 2
bash: fg: 2: no such job
root@rishardkogengar:/home/rkogengar# fg 3
dd if=/dev/zero of=/dev/null
^C86268489+0 records in
86268488+0 records out
44169465856 bytes (44 GB, 41 GiB) copied, 58.8239 s, 751 MB/s

root@rishardkogengar:/home/rkogengar# fg 4
sleep 7200
```

```
rkogengar@rishardkogengar:~ - top
rkogengar@rishardkogengar:/home/rkogengar
rkogengar@rishardkogengar:~ - top
```

```
top - 18:02:20 up 6 min, 4 users, load average: 0.24, 0.26, 0.13
Tasks: 260 total, 2 running, 258 sleeping, 0 stopped, 0 zombie
%Cpu(s): 5.4 us, 6.8 sy, 0.0 ni, 87.4 id, 0.0 wa, 0.3 hi, 0.1 si, 0.0 st
MiB Mem : 3909.0 total, 1304.7 free, 1425.3 used, 1418.4 buff/cache
MiB Swap: 4040.0 total, 4040.0 free, 0.0 used. 2483.7 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
3928	rkogeng+	20	0	226848	1776	1776	R	99.7	0.0	0:09.91	dd
2108	rkogeng+	20	0	4908712	309436	122288	S	2.3	7.7	0:03.74	gnome-shell
3224	rkogeng+	20	0	3029364	303952	96260	S	2.0	7.6	0:02.81	ptxix
123	root	20	0	0	0	0	I	0.3	0.0	0:00.11	kworker/u18:2-events_unbo
1	root	20	0	49192	41172	10248	S	0.0	1.0	0:01.47	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
9	root	20	0	0	0	0	I	0.0	0.0	0:00.07	kworker/0:1-events
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highp
11	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/u16:0-events_unbo
12	root	20	0	0	0	0	I	0.0	0.0	0:00.02	kworker/u16:1-netns
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
16	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
17	root	20	0	0	0	0	S	0.0	0.0	0:00.00	ksoftirqd/0
18	root	20	0	0	0	0	I	0.0	0.0	0:00.06	rcu_preempt
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_wo
20	root	20	0	0	0	0	S	0.0	0.0	0:00.08	rcu_exp_gp_kthread_worker
21	root	rt	0	0	0	0	S	0.0	0.0	0:00.01	migration/0
22	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
23	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
24	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1

Управление заданиями

```
rkogengar@rishardkogengar:~ - top
rkogengar@rishardkogengar:/home/rkogengar
rkogengar@rishardkogengar:~ - top

top - 18:02:41 up 6 min,  4 users,  load average: 0.66, 0.36, 0.16
Tasks: 260 total,  1 running, 259 sleeping,  0 stopped,  0 zombie
%Cpu(s):  5.0 us,  6.5 sy,  0.1 ni, 88.3 id,  0.0 wa,  0.2 hi,  0.0 si,  0.0 st
MiB Mem : 3909.0 total, 1290.4 free, 1439.3 used, 1418.8 buff/cache
MiB Swap: 4040.0 total, 4040.0 free,  0.0 used, 2469.7 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 3224 rkogeng+  20   0 3029364 323536 96260 S   2.4   8.1   0:03.09 ptysis
 2108 rkogeng+  20   0 4912024 309820 122288 S   0.6   7.7   0:03.93 gnome-shell
    1 root      20   0  49192   41172  10248 S   0.3   1.0   0:01.50 systemd
  195 root      20   0          0         0      0 I   0.3   0.0   0:00.08 kworker/u18:3-events_unbo+
    2 root      20   0          0         0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root      20   0          0         0      0 S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root        0 -20          0         0      0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root        0 -20          0         0      0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root        0 -20          0         0      0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
    7 root        0 -20          0         0      0 I   0.0   0.0   0:00.00 kworker/R-netns
    9 root      20   0          0         0      0 I   0.0   0.0   0:00.07 kworker/0:1-events
   10 root        0 -20          0         0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_high+
   11 root      20   0          0         0      0 I   0.0   0.0   0:00.00 kworker/u16:0-events_unbo+
   12 root      20   0          0         0      0 I   0.0   0.0   0:00.03 kworker/u16:1-netns
   13 root        0 -20          0         0      0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
   14 root      20   0          0         0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
   15 root      20   0          0         0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
   16 root      20   0          0         0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
   17 root      20   0          0         0      0 S   0.0   0.0   0:00.00 ksoftirqd/0
   18 root      20   0          0         0      0 I   0.0   0.0   0:00.07 rcu_preempt
   19 root      20   0          0         0      0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_wo+
   20 root      20   0          0         0      0 S   0.0   0.0   0:00.08 rcu_exp_gp_kthread_worker
   21 root       rt   0          0         0      0 S   0.0   0.0   0:00.01 migration/0
   22 root     -51   0          0         0      0 S   0.0   0.0   0:00.00 idle_inject/0
   23 root      20   0          0         0      0 S   0.0   0.0   0:00.00 cpuhp/0
   24 root      20   0          0         0      0 S   0.0   0.0   0:00.00 cpuhp/1
   25 root     -51   0          0         0      0 S   0.0   0.0   0:00.00 idle_inject/1
```

```
root@rishardkogengar:/home/rkogengar#
root@rishardkogengar:/home/rkogengar# dd if=/dev/zero of=/dev/null &
[1] 4485
root@rishardkogengar:/home/rkogengar# dd if=/dev/zero of=/dev/null &
[2] 4497
root@rishardkogengar:/home/rkogengar# dd if=/dev/zero of=/dev/null &
[3] 4499
root@rishardkogengar:/home/rkogengar# ps aux | grep dd
root          2  0.0  0.0      0      0 ?        S   17:56   0:00 [kthreadd]
root         91  0.0  0.0      0      0 ?        I<  17:56   0:00 [kworker/R-ipv6_addrconf]
root        1151  0.0  0.0 512956 3012 ?        Sl  17:56   0:00 /usr/sbin/VBoxService --pidfile
/var/run/vboxadd-service.sh
rkogeng+    2510  0.0  0.6 962680 25188 ?        Ssl 17:56   0:00 /usr/libexec/evolution-addressb
ook-factory
root        4485 99.4  0.0 226848 1900 pts/0    R   18:06   0:17 dd if=/dev/zero of=/dev/null
root        4497 99.2  0.0 226848 1756 pts/0    R   18:06   0:12 dd if=/dev/zero of=/dev/null
root        4499 98.5  0.0 226848 1872 pts/0    R   18:06   0:10 dd if=/dev/zero of=/dev/null
root        4535  0.0  0.0 227688 2176 pts/0    S+  18:06   0:00 grep --color=auto dd
root@rishardkogengar:/home/rkogengar#
```

Рис. 4: Отображение процессов dd

```
958 ?      Ssl  0:00 /usr/bin/python3 -sP /usr/sbin/firewalld --nofork --nopid
1149 ?     Sl   0:00 /usr/bin/VBoxDRMClient
1151 ?     Sl   0:00 /usr/sbin/VBoxService --pidfile /var/run/vboxadd-service.sh
--
2453 ?     Ssl  0:00 \_ /usr/libexec/goa-identity-service
2456 ?     Ssl  0:00 \_ /usr/libexec/evolution-calendar-factory
2469 ?     Ssl  0:00 \_ /usr/libexec/gvfs-mtp-volume-monitor
2483 ?     Ssl  0:00 \_ /usr/libexec/gvfs-gphoto2-volume-monitor
2505 ?     Ssl  0:00 \_ /usr/libexec/gvfs-goa-volume-monitor
```



```
rkogengar@rishardkogengar:~$  
rkogengar@rishardkogengar:~$ su  
Password:  
root@rishardkogengar:/home/rkogengar# dd if=/dev/zero of=/dev/null &  
[1] 5235  
root@rishardkogengar:/home/rkogengar# dd if=/dev/zero of=/dev/null &  
[2] 5247  
root@rishardkogengar:/home/rkogengar# dd if=/dev/zero of=/dev/null &  
[3] 5249  
root@rishardkogengar:/home/rkogengar# renice -n 5 5235  
5235 (process ID) old priority 0, new priority 5  
root@rishardkogengar:/home/rkogengar# renice -n 15 5235  
5235 (process ID) old priority 5, new priority 15  
root@rishardkogengar:/home/rkogengar# killall dd  
[1] Terminated dd if=/dev/zero of=/dev/null  
[2]- Terminated dd if=/dev/zero of=/dev/null  
[3]+ Terminated dd if=/dev/zero of=/dev/null  
root@rishardkogengar:/home/rkogengar#
```

Рис. 6: Работа с приоритетами dd

```
root@rishardkogengar:/home/rkogengar#  
root@rishardkogengar:/home/rkogengar# yes > /dev/null &  
[1] 5462  
root@rishardkogengar:/home/rkogengar# yes > /dev/null  
^Z  
[2]+  Stopped                  yes > /dev/null  
root@rishardkogengar:/home/rkogengar# fg 2  
yes > /dev/null  
^C  
root@rishardkogengar:/home/rkogengar# █
```

Рис. 7: yes без перенаправления

```
root@rishardkogengar:/home/rkogengar#  
root@rishardkogengar:/home/rkogengar# jobs  
[1]+  Running                  yes > /dev/null &  
root@rishardkogengar:/home/rkogengar# fg 1  
yes > /dev/null  
^Z  
[1]+  Stopped                  yes > /dev/null  
root@rishardkogengar:/home/rkogengar# bg 1  
[1]+ yes > /dev/null &  
root@rishardkogengar:/home/rkogengar# jobs  
[1]+  Running                  yes > /dev/null &  
root@rishardkogengar:/home/rkogengar# nohup yes > /dev/null &  
[2] 6121  
nohup: ignoring input and redirecting stderr to stdout  
root@rishardkogengar:/home/rkogengar# █
```

Рис. 8: Управление заданиями

Управление процессами с программой yes

```
rkogengar@rishardkogengar:/home/rkogengar - top
top - 18:20:10 up 24 min, 5 users, load average: 1.87, 1.25, 0.72
Tasks: 266 total, 4 running, 262 sleeping, 0 stopped, 0 zombie
%Cpu(s): 19.0 us, 33.3 sy, 0.0 ni, 47.6 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3909.0 total, 1398.0 free, 1319.0 used, 1431.8 buff/cache
MiB Swap: 4040.0 total, 4040.0 free, 0.0 used, 2590.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
5462	root	20	0	226820	1748	1748	R	90.9	0.0	6:50.10	yes
6121	root	20	0	226820	1732	1732	R	81.8	0.0	1:48.54	yes
1	root	20	0	49192	41044	10248	S	0.0	1.0	0:02.46	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
9	root	20	0	0	0	0	I	0.0	0.0	0:00.08	kworker/0:1-cgroup_destroy
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highp
11	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/u16:0-events_unbo
12	root	20	0	0	0	0	I	0.0	0.0	0:00.10	kworker/u16:1-netns
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
16	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
17	root	20	0	0	0	0	S	0.0	0.0	0:00.01	ksoftirqd/0
18	root	20	0	0	0	0	I	0.0	0.0	0:00.17	rcu_preempt
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_wo
20	root	20	0	0	0	0	S	0.0	0.0	0:00.12	rcu_exp_gp_kthread_worker
21	root	rt	0	0	0	0	S	0.0	0.0	0:00.01	migration/0
22	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
23	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
24	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
25	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
26	root	rt	0	0	0	0	S	0.0	0.0	0:00.22	migration/1
27	root	20	0	0	0	0	S	0.0	0.0	0:00.01	ksoftirqd/1
29	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/1:0H-events_highp

Управление процессами с программой yes

```
root@rishardkogengar:/home/rkogengar# yes > /dev/null &
[1] 6582
root@rishardkogengar:/home/rkogengar# yes > /dev/null &
[2] 6584
root@rishardkogengar:/home/rkogengar# yes > /dev/null &
[3] 6587
root@rishardkogengar:/home/rkogengar# kill 6582
[1] Terminated yes > /dev/null
root@rishardkogengar:/home/rkogengar# fg 2
yes > /dev/null
^C
root@rishardkogengar:/home/rkogengar# kill -1 5462
root@rishardkogengar:/home/rkogengar# kill -1 6582
bash: kill: (6582) - No such process
root@rishardkogengar:/home/rkogengar# yes > /dev/null &
[4] 6797
root@rishardkogengar:/home/rkogengar# yes > /dev/null &
[5] 6799
root@rishardkogengar:/home/rkogengar# yes > /dev/null &
[6] 6811
root@rishardkogengar:/home/rkogengar# killall yes
[4] Terminated yes > /dev/null
[3] Terminated yes > /dev/null
[5]- Terminated yes > /dev/null
[6]+ Terminated yes > /dev/null
root@rishardkogengar:/home/rkogengar#
```

Управление процессами с программой yes

```
root@rishardkogengar:/home/rkogengar# yes > /dev/null &
[1] 6872
root@rishardkogengar:/home/rkogengar# nice -n 5 yes > /dev/null &
[2] 6895
root@rishardkogengar:/home/rkogengar# ps -l
```

F	S	UID	PID	PPID	C	PRI	NI	ADDR	SZ	WCHAN	TTY	TIME	CMD
4	S	0	6419	6384	0	80	0	- 58153	do_wai	pts/1		00:00:00	su
4	S	0	6430	6419	0	80	0	- 57575	do_wai	pts/1		00:00:00	bash
4	R	0	6872	6430	99	80	0	- 56705	-	pts/1		00:00:11	yes
4	R	0	6895	6430	98	85	5	- 56705	-	pts/1		00:00:04	yes
4	R	0	6897	6430	0	80	0	- 57682	-	pts/1		00:00:00	ps

```
root@rishardkogengar:/home/rkogengar# renice -n 5 6872
6872 (process ID) old priority 0, new priority 5
root@rishardkogengar:/home/rkogengar# ps -l
```

F	S	UID	PID	PPID	C	PRI	NI	ADDR	SZ	WCHAN	TTY	TIME	CMD
4	S	0	6419	6384	0	80	0	- 58153	do_wai	pts/1		00:00:00	su
4	S	0	6430	6419	0	80	0	- 57575	do_wai	pts/1		00:00:00	bash
4	R	0	6872	6430	99	85	5	- 56705	-	pts/1		00:00:42	yes
4	R	0	6895	6430	99	85	5	- 56705	-	pts/1		00:00:34	yes
4	R	0	6965	6430	0	80	0	- 57682	-	pts/1		00:00:00	ps

```
root@rishardkogengar:/home/rkogengar# killall yes
[1]- Terminated                  yes > /dev/null
[2]+ Terminated                  nice -n 5 yes > /dev/null
root@rishardkogengar:/home/rkogengar#
```

Рис. 11: Выравнивание приоритетов

Итоги работы

В ходе лабораторной работы были освоены основные приёмы управления процессами в Linux